A League of Legends Gameplay

This section introduces the core game mechanics and player roles, necessary for the analysis in this thesis. The focus is on match structure, key objectives, roles, and the professional competitive context.

A.1 Match Structure & Objective

The standard game mode examined in this thesis is played by two teams, each consisting of five players. Each player selects an individual champion, all having four abilities. League of Legends has currently more than 170 available unique champions, each with a distinct playstyle, item-builds and strengths. This creates a large variety of potential team compositions and interactions with opposing teams. Especially in professional play the composition of champions is important, with every professional draft having an essential identity and playstyle.

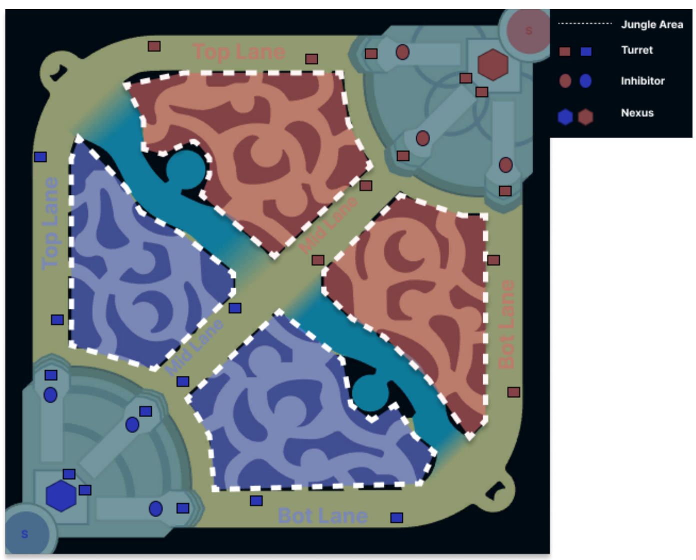

Fig. 2. Illustration, based on "Summoners Rift" map from League of Legends. Teams progress through lanes and defensive structures to reach and destroy the opposing Nexus (denoted by hexagon), the ultimate win condition of a match. See supplementary material for details. Original map is intellectual property of Riot Games[16].

The map, as seen in Figure 2, is divided into two sides (red and blue) separated by a river, with three lanes, Top, Mid, and Bot, connecting the teams bases. Each lane contains three turrets and one inhibitor per side. In Figure 2, the turrets are denoted by rectangles and inhibitors by circles, where the color of the structure corresponds to the team it belongs to. Each lane is progressed by destroying the structures in a lane, where destruction grants gold directly, and pressure indirectly. The structures can only be destroyed in sequential order, from furthest to closest to the base of the team the structure belongs to, meaning the inhibitor can only be destroyed after the three turrets of the team, on that specific lane, have already been destroyed. Turrets deal significant damage to enemy units, providing defensive zones for teams. As seen in Figure 2, the lanes are labeled "Top Lane", "Mid Lane" and "Bot Lane", which is short for Bottom Lane.

The four separate areas in between the lanes, marked by the dotted lines in Figure 2, are called the Jungle area, where the color again denotes the corresponding team. The jungle contains neutral monsters that spawn at regular intervals, with epic monsters/neutral objectives spawning at fixed times in the river pits and providing team-wide advantages when defeated. Lastly minions spawn in each lane periodically throughout the game.

The term "farming" refers to killing the minions or normal monster, depending on the role, and the term "farm" refers to the amount of monsters or minions killed, being measured by the creep score. Both monsters and minions grant gold and experience when slain, representing the most consistent form of gold generation.

Gold enables item purchases, which in turn increase champion power. Gold can be obtained by farming either minions or monsters, as well as by destroying turrets. Additionally, killing opposing players, as well as assisting in the kill, grants gold. The amount of gold for a kill varies depending on game state and specific player performance, but the kill will always grant more gold then the assist.

Another concept essential to League of Legends is vision control, shown generally through the aggregated metric of visionscore. Vision control describes the amount of vision a team has on the map, as well as the amount of vision destroyed of the opposing team. To obtain additional vision, different types of "wards" can be used, which differ in amount of vision granted and time for which vision is granted. These wards can be destroyed through multiple interactions, including items, runes and other game mechanics.

The game is generally defined into three phases, commonly agreed upon in the community:

1. **Early Game / Laning Phase ($\sim 0 - 15$ min.):** Focus on farming and lane control by generating advantages, with occasional minor skirmishes
2. **Mid Game / End of Laning Phase ($\sim 15 - 20$ min.):** Focus around contesting neutral objectives, team rotations and the resulting team fights
3. **Late Game ($\sim 20+$ min.):** Fights and neutral objectives become critical, and have the ability to decide the game outcome, with most fights being team fights.

Lastly, the ultimate goal of a match is to destroy the enemy Nexus, located deep in each teams base and denoted by a respectively colored hexagon in Figure 2. Nexus destruction requires first eliminating at least one inhibitor, as well as destruction of the two turrets defending the Nexus.

A.2 Overview of the Roles and Team Compositions

In this section we will build on the foundation laid by the previous section, by giving an overview of the specific roles and the basics of team compositions. As explained in Section A.1, as well as seen on Figure 2, the map is divided into four general areas per side, the three lanes and the jungle area. In respect to these areas, the following five roles have been established

1. Top(-Laner)
2. Mid(-Laner)
3. Bot/Bottom(-Laner)
4. Support
5. Jungler

The role names generally indicate the primary map position, in which a player operates during the early stages of the game. For example, the Jungler mainly resides in the jungle area and the Mid Laner in the mid lane. While each role is associated with typical responsibilities, there usually is no single fixed champion archetype assigned to a role, as champion selection depends on team strategy, synergy considerations, and tactical objectives. The following will give a short explanation and overview of all of the five roles, as agreed upon in the community[9, 11, 3]:

– **Top(-Laner):** Usual champions are characterized by durability, and are good at operating in isolation. Contributing through individual map pressure or frontline presence in team fights, depending on team composition and opponent.
– **Jungler:** Operates between lanes and gains resources through periodically farming neutral camps. Characterized by map pressure on all lanes and the responsibility of contesting neutral objectives. Champion type is diverse, but typically provides some aspects of mobility, damage output and crowd control utility.
– **Mid(-Laner):** Plays in the central lane and typically uses champions providing some aspects of play-making ability, team fighting utility or map mobility. Assisting in other lanes and neutral objectives, due to central position.
– **Bot(-Laner) / Attack Damage Carry (ADC):** Typically Marksman-type champions having high amounts sustained ranged damage output, with low defensive stats. Paired with support role in the Bot-Lane, playing key role in team fights and scaling with gold and items.
– **Support:** Assists the ADC in the early game, with the ability to assist other lanes as needed. Provides vision, protection utility and crowd control depending on the specific champion played.

Team compositions are primarily shaped by a team's strategic objectives and intended win conditions. Champions are selected to create synergistic interactions and to support specific tactical approaches, rather than to fulfill rigid archetypal templates. Many compositions revolve around particular win conditions, such as prioritizing utility for a strong central champion, focusing on coordinated team fighting, or emphasizing early or late-game advantages.

A.3 Context of Professional Competition

This study exclusively uses professional League of Legends match data, rather than ranked or casual match data. The primary reason for this choice, is data quality and behavioral consistency. Professional matches are played in a highly controlled competitive environment, in which all participants are incentivized to win and perform optimally, in terms of their role and responsibilities. We believe that non-professional matches have substantially higher noise, due to non-competitive behavior, unstructured team fighting or other factors.

Professional play also ensures a consistently high mechanical and strategic skill level across all roles. Thus, observed performance differences are more likely to reflect meaningful strategic and executional variation, rather than basic skill gaps or misunderstanding of role responsibilities. This makes role-based impact comparisons, such as between ADC and Jungle, more valid and interpretable.

Another important factor is draft quality and role clarity. Professional teams use structured drafting processes, supported by coaching staff, scouting, and preparation. Champion selections are therefore intentional and strategically coherent, rather than preference driven or random. This reduces uncontrolled variance from poor compositions and increases the likelihood, that each role is placed in a composition designed to function properly, which strengthens internal validity when attributing outcome effects to specific roles.

Additionally, objective control, vision setup, lane assignments, and resource allocation are executed according to established strategic principles, which creates more stable and comparable game states across matches, making it easier to model how ADC and Jungle contributions translate into win probability. In lower tier play, chaotic decision making and inconsistent macro behavior introduce confounding variation, which weakens causal inference.

Lastly, the professional context helps satisfy a key assumption for causal analysis: that observed behavior reflects genuine competitive intent rather than arbitrary or low effort play.

For these reasons, higher behavioral reliability, strategic structure, draft quality, role clarity, statistical accuracy, and incentive alignment, professional League of Legends matches provide a suitable dataset for estimating the causal impact of ADC and Jungle roles on match outcomes.

A.4 Role Specific Performance Definition

As this thesis aims to investigate the impact of player roles on match outcomes, with a particular focus on the Jungle and ADC roles, it is necessary to define what constitutes role-specific impact. Sharpe et al. [18] discusses that in League of Legends, player performance cannot be fully captured by single quantitative metrics alone, as the effectiveness of a player depends on how well they execute their responsibilities inherent to their assigned role. Therefore, this section examines the core responsibilities, decision making priorities, and strategic functions of the ADC and Jungle roles. By understanding these roles in the context of their typical in-game duties, we can establish a framework for evaluating player impact, based on the successful fulfillment of the responsibilities, rather than relying solely on raw statistics such as kills or gold earned.

The specification of role responsibilities is based on domain knowledge as well as community blogs or post [9, 11, 3], as there is no concrete definition by Riot Games.

The ADC or the Bot-Lane role is typically occupied by a marksman champion, as mentioned in Section A.2. The marksman champion type is characterized by dealing high and consistent amounts of damage from range, with low defensive stats. Additionally, guides and blogs similar to [9, 3] outline the characteristic of scaling with gold and items and thus the responsibility of farming efficiently to faster achieve the power accompanied with the items. As explained before, starting in the mid-game team fights become increasingly more important, as does the presence of the ADC in team fights. The ADC role is expected to be the main consistent damage source, which results in the ADC being the highest priority target for opposing teams, as the longer an ADC is alive the more damage they could potentially deal. Thus, the last crucial aspect of the ADC role is positioning. Conclusively, we identify the responsibilities of the ADC role as follows:

- Efficient farming and gold acquisition
- Providing sustained ranged damage output, especially in team fights
- Maintaining safe positioning to maximize damage up-time and survivability

Continuing with the Jungle role, there exists a wide range of viable champions and no single dominant archetype is universally preferred. The Jungler differs from the lane roles in not occupying a fixed lane, but instead operating primarily in the jungle area between lanes. This structural difference provides greater movement flexibility across the map.

This flexibility is reflected in the Jungler's responsibilities, which include creating advantages for teammates through coordinated lane pressure, commonly referred to as ganking. A central aspect of the role, is neutral objective control, including monsters that provide team wide benefits, as also emphasized in role descriptions [9, 11].

As the role must balance resource collection with map impact, Junglers are required to make continuous decisions between farming neutral camps, applying pressure to lanes, and securing objectives. Efficient Jungle pathing and timing of map movements are therefore considered core skills associated with the role. Based on these commonly described characteristics, the Jungle role responsibilities can be summarized as:

- Applying map pressure through ganks and lane assistance
- Securing and contesting neutral objectives
- Managing jungle farm efficiently through pathing and timing decisions

The defined responsibilities of both roles guide the extraction of role-specific metrics and evaluation of player impact in the later analysis.

B Causal Graph

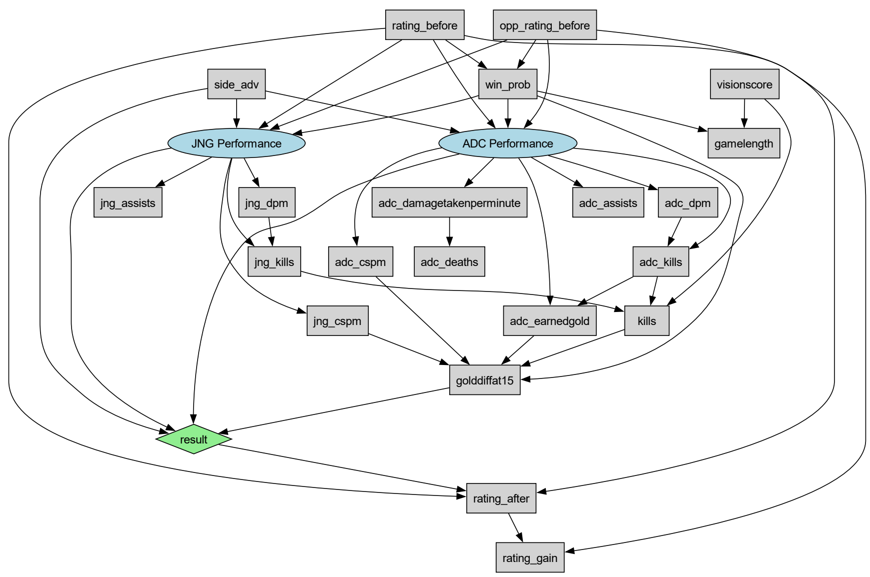

Fig. 3. Full causal model including ADC and JNG latent performance variables. Ellipses denote latent variables, boxes denote observed indicators. A directed edge $X \to Y$ indicates that X is a cause of Y.

Table 4. Sufficient adjustment sets corresponding to the causal graph shown in Fig. 2. These sets *block* non-causal associational influences from treatment to the outcome variable. Adjusting for these variables allows for estimation of true causal impact from treatment to the outcome.

Effect	Treatment	Outcome	Adjustment set
τ_1	T_{ADC}	Y_{win}	$\{\texttt{side\_adv}, \texttt{win\_prob}, T_{\text{JNG}}\}$
τ_2	T_{JNG}	Y_{win}	$\{\texttt{side\_adv}, \texttt{win\_prob}, T_{\text{ADC}}\}$
τ_3	T_{ADC}	Y_{gain}	$\{\texttt{side\_adv}, \texttt{win\_prob}, \texttt{rating\_before}, \texttt{opp\_rating\_before}\}$
τ_4	T_{JNG}	Y_{gain}	$\{\texttt{side\_adv}, \texttt{win\_prob}, \texttt{rating\_before}, \texttt{opp\_rating\_before}\}$

C Details on Refutation Tests

1. **Placebo Treatment:** Replaces the treatment with a random variable. The estimated effect should collapse to approximately zero, indicating that the original effect is linked to the treatment rather than statistical artefacts.
2. **Random Common Cause:** Introduces a random variable as an additional confounder. The estimated effect should remain stable, indicating that the estimate is robust to irrelevant covariates.
3. **Data Subsample:** Re-estimates the effect on random subsets of the dataset. The estimated effect should remain consistent across subsamples, indicating that the result is not driven by a subset of matches.

D Details on Sensitivity Analysis

The methods of Cinelli and Hazlet [6] propose analysing partial R^2 measures for the results, estimating how large an unobserved confounder would have to be to overturn the results. The method assumes for an unobserved confounder U to exist. We then calculate the following measures:

1. **Treatment Association:** $R^2_{T_r \sim U|X}$ = proportion of the residual variance in the treatment T_r explained by U after controlling for X
2. **Outcome Association:** $R^2_{Y_y \sim U|T_r,X}$ = proportion of the residual variance in the outcome Y explained by U after adjusting for T and X

where X is the set of observed confounders. Using these measures a robustness value (RV_q) is defined, which describes how strong the confounder's partial R^2 would have to be, equal for both treatment and outcome association, to reduce the effect by a proportion q. In simpler terms, $RV_{q=1}$ is the minimum strength of confounding necessary to fully explain away the estimated effect. In this work, we only consider $RV_{q=1}$, which is referred to as simply RV, and $RV_{\alpha=0.05}$, which describes the strength necessary to make the result statistically insignificant for $\alpha = 0.05$.